Missing citations in "Full support of the zero-temperature Sherrington–Kirkpatrick order parameter"

October 13, 2026

Summary of findings

Each row is a citation the preprint is missing: earlier work that already contains part of the preprint, which the preprint does not cite where that part appears. "Cited?" names the earlier work and says what, if anything, the preprint cites it for. The exception is a row flagged *overstated novelty*: there the preprint cites the work, but its abstract or framing presents the content as new. Terms are defined in Section 1; centrality is a best-effort judgment. Locations use the preprint's own numbering and printed pages.

ID	Relationship	Centrality	Technique or result	In the preprint	Cited?
F1	*same result*	main result	Full support on $[0,1)$ of the zero-temperature order parameter, with strict increase and (1.6)	Abstract (p. 1); Thm 1.1 (p. 2)	H.-B. Chen 2026 [1], first in its aiXiv version [2]: cited after Thm 1.1 (p. 2), not in the abstract; **overstated novelty**
F2	*same technique*	central	Gap exclusion by the endpoint contact relations against a positive third derivative on the gap	§1.2 and Fig. 1 (p. 4); Lemma 2.3 (p. 7); proof of Thm 1.1 (p. 20)	Zhou 2025 [3]: cited for other content; Lopatto 2026 [4]: Prop. 9.2 credited elsewhere (§1.1, p. 3), Lemma 2.5 cited for other content; H.-B. Chen 2026 [1]: cited for other content; **attribution gap**
F3	*same result*	supporting	The Cole–Hopf slope invariant as sign inequalities for the backward drift	Lemma 3.2 (p. 11)	Lopatto 2026 [4]: credited elsewhere (§1.1, p. 3); H.-B. Chen 2026 [1]: credited elsewhere in part
F4	*same result*	supporting	Monotonicity of the ratio of the forward score to the drift, invariant at jumps	Lemma 3.3 (pp. 12–13)	Zhou 2026 [5]: not cited; **attribution gap**

ID	Relationship	Centrality	Technique or result	In the preprint	Cited?
F5	*same result*	supporting	Exclusion of a terminal plateau by a square-root boundary layer	Lemma 6.3 (pp. 19–20)	H.-B. Chen 2026 [1], first in its aiXiv version [2]: cited for other content; Auffinger–Chen–Zeng 2017 [6]: credited for the conclusion; El Alaoui–Montanari–Sellke 2020 [7]: credited in part; **attribution gap**
F6	*same result*	supporting	The origin lies in the support: no initial interval of zero mass	Proof of Thm 1.1, initial-gap case (p. 20)	Chen–Handschy–Lerman 2016 [8]: cited for other content; Auffinger–Chen 2013 [9]: cited for other content; **attribution gap**
F7	*same result*	peripheral	Full support verifies the no-overlap-gap assumption, giving an ϵ-optimal algorithm	Cor. 7.3 (p. 25); announced on pp. 2–3	H.-B. Chen 2026 [1]: cited for other content

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1 About this document

The preprint. The preprint is *Full support of the zero-temperature Sherrington–Kirkpatrick order parameter*, dated September 27, 2026, with author line "OpenAI" [10]. It is one of the preprints of the OpenAI Math Release. It proves that every minimizer of the zero-temperature Parisi functional for the pure, zero-field Sherrington–Kirkpatrick model has full support on $[0, 1)$, and that the minimizing order parameter has a smooth, strictly positive density on $(0, 1)$. Section 2 summarizes the paper and fixes the notation used throughout this document. The preprint cites no other preprint from the same release.

What this document reports. Each finding is a citation the preprint is missing: part of the preprint's mathematics (a technique, construction, estimate, proof route or result) that an identifiable earlier paper already contains, where the preprint does not cite that paper for it where it is used. "Earlier" means publicly available before the preprint's date. The one exception, overstated novelty, is described below. Not reported: textbook material in its textbook form; weaker versions of intermediate statements; and, for content the preprint uses, cases where it cites a later source that covers that content rather than the original. An earlier result on the same problem is reported even when later work that the preprint cites implies it. Findings are ordered by centrality, then by relationship in the order listed below, then by position in the preprint.

Relationships. A finding's relationship is the one between the preprint and the first earlier work listed for it (the principal), the work with the highest relationship in the order below; when several hold that relationship, however closely each matches, the one the others credit or build on, otherwise the earliest. Any other works listed relate to the same content more weakly or in part; the finding says how, including which work originated the idea when that is not the first.

same result. An earlier paper, or several earlier papers together, prove essentially the same statement.

same technique. The earlier technique, construction or theory is reused essentially intact, possibly in a new setting or with new ingredients around it.

adapted. The earlier technique's structure is followed, with a change at its core: a different object, or a key step replaced.

proof shortcut. A published result would replace or substantially shorten a large part of the argument.

same route. An earlier paper proposed the proof strategy, even heuristically, and the preprint carries it out.

fragment. One identifiable piece of the earlier technique is reused.

weaker result. An earlier paper proves a weaker, special-case or partial-range version of one of the preprint's main results, or of a statement it presents as a significant contribution.

Centrality. Each finding also gets a best-effort judgment of how much of the preprint's main argument depends on it: *main result* (reserved for earlier work that covers the main result itself, or most of it), *central*, *supporting* or *peripheral*. An earlier weaker result is never labelled main result. Each finding states the chain of results it feeds, so the basis for the judgment is visible.

Citations. The "Cited?" column of the summary names the earlier work and says what the preprint cites for the content in question: *not cited* (the work is not in the bibliography, and nothing the preprint cites credits it for this content), *cited for other content*, *credited elsewhere* (credited for this content, directly or through a cited location that credits it, but not where it is used, in a way that an expert reader who doesn't know the earlier work might not connect with it), or *credited in part*. A citation counts only when it is *for that specific content*; citing the same paper nearby for a different result does not count. Content the preprint credits where it is used, directly or through a cited location that itself credits the earlier work, is not reported, except when its novelty is overstated (below). For overstated novelty, the column says where the preprint does cite the work.

Flags.

Attribution gap. Marks the most serious missing citations: the relationship is same technique, adapted or same result; the centrality is main result, central or supporting; and the preprint cites the principal earlier work for this content nowhere.

Overstated novelty. The preprint does cite the earlier work for this content, but its abstract, a statement of its significant contributions, or the framing of a main result presents that content as new, explicitly or in the way a reader would reasonably take it. This is the one case in which content the preprint credits is reported.

How to check a claim. Locations in the preprint use its section, equation and theorem numbers and the page numbers printed in its PDF. Locations in other sources give the theorem, equation and printed page of the version listed in the bibliography; where an entry lists both a journal and an arXiv version, they refer to the arXiv version. Each finding ends with a short checklist of the passages to put side by side.

Provenance. This document was generated by LLM agents (Claude Opus 5.5) running the precedence-review skill (jnshi/math-lineage, commit `13ccc19`; parts of the review ran under the earlier versions `d1ea740` and `3808a6c`), makes no claim of completeness, and has not been vetted by a human. It was run by Jonathan Shi.

2 The preprint in brief, and notation

Model and variational problem. The pure, zero-field Sherrington–Kirkpatrick Hamiltonian is $H_n(\sigma) = n^{-1/2}\sum_{i<j} J_{ij}\sigma_i\sigma_j$, normalized so that the covariance is $\xi(t) = t^2/2$ and $\xi'' \equiv 1$. The class $\mathcal{U}$ consists of nonnegative, nondecreasing, right-continuous $\gamma : [0,1) \to [0,\infty)$ with $\int_0^1 \gamma < \infty$, and μ_γ is the Stieltjes measure with $\mu_\gamma([0,t]) = \gamma(t)$. The Parisi value function Φ_γ solves

$$\partial_t \Phi = -\tfrac{1}{2}(\Phi_{xx} + \gamma(t)\Phi_x^2), \qquad \Phi(1,x) = |x|,$$

(1.1)–(1.2), and the zero-temperature Parisi functional is $\mathcal{P}(\gamma) = \Phi_\gamma(0,0) - \frac{1}{2}\int_0^1 t\gamma(t)\,dt$ (1.3). Its minimum over $\mathcal{U}$ is the limiting ground-state energy per spin (1.4).

The optimal diffusion and the self-consistency function. Write $u = \Phi_{\gamma,x}$ and let X solve $dX_t = \gamma(t)u(t,X_t)\,dt + dB_t$, $X_0 = 0$ (1.5). Set

$$q(t) = \mathbb{E}[u(t,X_t)^2], \qquad q'(t) = \mathbb{E}[\Phi_{\gamma,xx}(t,X_t)^2]$$

(2.12), Lemma 2.2. The sources below write Γ for this function, and m for the value of a constant stretch of the order parameter.

Main theorem. Theorem 1.1: if $\gamma \in \mathcal{U}$ minimizes $\mathcal{P}$, then $\operatorname{supp}\mu_\gamma = [0,1)$, and $q(t) = t$ and $\mathbb{E}[\Phi_{\gamma,xx}(t,X_t)^2] = 1$ for $0 \le t < 1$. Corollary 6.4 adds that γ is smooth on $[0,1)$ with $\gamma(0) = 0$ and $\gamma' > 0$ on $(0,1)$.

Contact conditions. For a minimizer, the contact potential $G(s) = \int_s^1 (q(t) - t)\,dt$ is nonnegative and vanishes on the support; at every interior support point, $q(s) = s$ and $q'(s) \le 1$ (Lemma 6.2, p. 19).

Notation on a constant interval. On an interval where $\gamma \equiv c > 0$, the preprint sets

$$r = c\Phi_x, \qquad v = r_x, \qquad w = r_{xx}, \qquad z = r_{xxx},$$

and proves

$$(q')' = c^{-2}\mathbb{E}[w^2 - 2v^3], \qquad (q')'' = c^{-2}\mathbb{E}[z^2 - 12vw^2 + 6v^4]$$

along the diffusion ((2.13)–(2.14), Lemma 2.3).

Proof route.

- Proposition 5.1: on every positive constant interval, $q''' \ge (1000c^2)^{-1}\mathbb{E}v^4 > 0$, so q' is strictly convex there.
- Interior gaps (proof of Theorem 1.1, p. 20): on a gap (a,b) with endpoints in the support, $q(a) = a$, $q(b) = b$ and $q'(a), q'(b) \le 1$. Strict convexity puts q' below its chord, hence below 1, on (a,b), so $q(b) - q(a) < b - a$, a contradiction (Figure 1, p. 4). Gaps on which $\gamma = 0$ are excluded by strict monotonicity of q'.
- A terminal plateau is excluded separately (Lemma 6.3, pp. 19–20).

The machinery behind Proposition 5.1. The preprint's §§3–4 work with a positive finite-step coefficient $c_1 \le \cdots \le c_N \le M$ and terminal datum $M^{-1}\log(2\cosh Mx)$ (§3, p. 8). They factor the density of X_t as $f = e^{c\Phi}g$, where g solves the forward heat equation between jumps, and put $s = -\partial_x \log g$ (§3, p. 8). With backward time τ and $H = r_\tau/r$:

- Lemma 3.2: on $x > 0$, $w \le 0$, $H \le 0$ and $H_x \ge 0$;
- Lemma 3.3: $s_x \ge 1/t$, $s_{xx} \le 0$ on $x > 0$, and $j := s_x - (s/r)v = r\,\partial_x(s/r) \ge 0$;
- §3.4 and Lemma 3.4: the variables $R = r^2$, $K = rs$, $b = -w/r$, $B = b - 2v = -2H$, $S = z + bv - 2Rb = 2rH_x$ and $L = -rs_{xx}$ are nonnegative, and $\mathbb{E}_f[\mathcal{D}F + (v + R - K)F] = 0$ for polynomials F, where $\mathcal{D} = r\partial_x$;
- Lemma 4.1 and Proposition 4.2: an exact identity over $\mathbb{Q}$ writes $z^2 - 12vw^2 + 6v^4$ as a sum of nonnegative terms plus such a divergence, giving $\mathbb{E}_f[z^2 - 12vw^2 + 6v^4] \ge 10^{-3}\mathbb{E}_f v^4$.

3 Findings

F1. Full support of the zero-temperature order parameter

Summary. H.-B. Chen's paper [1, Thm 1.3, p. 3] (arXiv, July 20, 2026) proves that the minimizer of the zero-temperature Parisi functional of the pure, zero-field SK model has full support on $[0, 1)$; the theorem first appeared in its aiXiv version, posted July 16, 2026 [2, Thm 1.2, p. 2], which Chen's paper credits [1, p. 1; App. C, p. 38]. The preprint cites Chen's paper for this theorem right after Theorem 1.1, p. 2, but its abstract (p. 1) opens "We prove that" and presents the theorem as new, naming neither Chen's paper nor an earlier proof. The relationship is *same result*: every assertion of Theorem 1.1 (full support, strict increase of γ, and (1.6)) is in Chen's paper, for the same model, class, functional and notion of support. Both proofs follow the plan of Chen's paper, first set out in its aiXiv version [2, pp. 2–3]: exclude each component of the complement of the support directly at zero temperature, using that γ is constant on it and the first-variation contact conditions (a gap between support points through the frame of finding F2, a terminal plateau through the boundary layer of finding F5). In the gap step, Chen's paper gets the contradiction from the crossing rule $\Gamma'' = 0 \Rightarrow \Gamma''' > 0$ [1, Prop. 4.2, p. 19] and the preprint from the unconditional bound $q''' \ge (1000c^2)^{-1}\mathbb{E}v^4$ of Proposition 5.1, a difference the preprint states (pp. 2–3).

Correspondence.

	Preprint	Chen [1]
Setting	$\xi(t) = t^2/2$; $\mathcal{U}$, $\mu_\gamma([0,t]) = \gamma(t)$; (1.2)–(1.3) (pp. 1–2)	$\xi(t) = t^2/2$; $\mathcal{U}$ (1.4), $\mathrm{d}\gamma([0,t]) = \gamma(t)$; (1.5)–(1.6) (p. 3)
Full support	every minimizer $\gamma \in \mathcal{U}$: $\operatorname{supp}\mu_\gamma = [0,1)$, and $\gamma(b) > \gamma(a)$ for $0 \le a < b < 1$ (Theorem 1.1, p. 2)	the minimizer $\gamma_\star$, unique by Chen, Handschy and Lerman [8, Thm 4, p. 8]: $\operatorname{supp}_{[0,1)}\nu_\star = [0,1)$, $\nu_\star = \mathrm{d}\gamma_\star$ (Thm 1.3, (1.8), p. 3), and $\gamma_\star(t) - \gamma_\star(s) = \nu_\star((s,t]) > 0$ for $0 \le s < t < 1$ (p. 4)
Saturated consistency	$\mathbb{E}[u_\gamma(t, X_t)^2] = t$, $\mathbb{E}[\Phi_{\gamma,xx}(t, X_t)^2] = 1$, $0 \le t < 1$ (1.6)	$\Gamma(t) = t$ (4.40), $\mathbb{E}\,\mathsf{C}(t, X_t)^2 = 1$ (4.41), $0 \le t < 1$, where $\mathsf{C} = u_{xx}$ and $u = \Phi_\gamma$ (proof of Prop. 4.6, p. 25)

How to check.

1. Theorem 1.1, p. 2 against [1, Thm 1.3, (1.8), p. 3], the display on p. 4 of Chen's paper, and [2, Thm 1.2, p. 2], whose (1.3) is the second identity in Chen's (1.8).
2. (1.6) against [1, (4.40)–(4.41), p. 25].
3. The preprint's abstract (p. 1) against its citation of Chen's paper after Theorem 1.1, p. 2.

F2. Excluding support gaps by the endpoint contact relations and a positive third derivative

Summary. Zhou excludes a gap (a, b) near the origin in the support of the near-critical SK Parisi measure by showing that the endpoint relations $\Gamma(a) = a$, $\Gamma(b) = b$, $\Gamma'(a), \Gamma'(b) \le 1$ [3, (7), p. 9] contradict $G(a, b) > 0$, which for short gaps he derives from a lower bound on $\partial_t^2 G(a, a+) = \Gamma'''(a+)/(3\Gamma'(a))$ and a bound on $\partial_t^3 G$ [3, Thm 5(2)–(3), Cor. 6, p. 8; pp. 14–15]. The preprint cites Zhou only for his interval-support theorem (§1.1, p. 3), and not where it uses his approach: §1.2, p. 4, Lemma 2.3, p. 7 (whose (2.14) is Zhou's Γ''' formula and sets the target of §§3–4) and the proof of Theorem 1.1, p. 20 (its initial-gap case is finding F6). The approach and its insight are Zhou's (Lopatto [4, p. 3] calls the general strategy closely related to Zhou's argument): on a gap the order parameter is constant, so Γ has explicit higher derivatives along the optimal diffusion, and $\Gamma''' > 0$ makes the trapezoid value $(b-a)(\Gamma'(a)+\Gamma'(b))/2$, at most $b-a$ by the slope relations, exceed $\int_a^b \Gamma' = b - a$. *same technique*: the frame is reused intact, and the integrated chord bound is Zhou's $G(a, b) > 0$; the steps differ only in the positivity the frame needs, which Proposition 5.1, p. 17 gives on every positive constant interval, where Zhou's bound is local and near-critical and the crossings of Lopatto and Chen 2026, which the preprint cites, are conditional. Lopatto has the same formulas [4, Lemma 2.5, p. 6], never cited, and the frame with such a crossing [4, Prop. 9.2, pp. 50–51], cited only through his Sections 3, 7–9 in the introduction (p. 3); Chen's paper [1, Prop. 4.3, pp. 22–23], cited for other results, adapts the frame at zero temperature, closing with a weighted identity instead of the chord (first in its aiXiv version of July 16, 2026 [2, Prop. 4.2, pp. 30–31]).

Correspondence.

	Preprint	Earlier work
Endpoint relations	$q(a) = a$, $q(b) = b$, $q'(a), q'(b) \le 1$ ((6.4), p. 19; p. 20)	Zhou (7), p. 9. Lopatto (9.32), p. 51: $h(a) = h(b) = 0$, $h'(a), h'(b) \le 0$, $h = \Gamma - s$. Chen 2026 (4.23)–(4.24), p. 22: $\Gamma(a) = a$, $\Gamma(b) = b$, $\Gamma'(a+) \le 1$, $\int_a^b (\Gamma - t) = 0$
Third derivative	$(q')'' = c^{-2}\mathbb{E}[z^2 - 12vw^2 + 6v^4]$ ((2.13)–(2.14), p. 7)	Zhou p. 15, with $8\beta^6 = (\xi'')^3$, here 1, and $m = c$: $\Gamma''' = 8\beta^6\mathbb{E}[((\partial_x^4\Phi)^2 - 12m\,\partial_x^2\Phi(\partial_x^3\Phi)^2 + 6m^2(\partial_x^2\Phi)^4)e^W]$. Lopatto (2.18), p. 6. Chen 2026 (3.38), p. 14: Γ'' only
Closing step	q' below its chord, hence below 1, on (a, b); integrating, $q(b) - q(a) < b - a$ (p. 20)	Zhou: $G(a, b) > 0$, $G(s, t) = (t - s)[\Gamma'(s) + \Gamma'(t)]/(\Gamma(t) - \Gamma(s)) - 2$ (pp. 8–9). Lopatto: $\max h' = \max\{h'(a), h'(b)\} \le 0$, $0 = \int_a^b h'$ (p. 51). Chen 2026: $\int_a^b (t - a)(b - t)\Gamma'' = 0$, so $\Gamma(b) - \Gamma(a) < (b - a)\Gamma'(a+)$ ((4.25)–(4.26), p. 23)

How to check.

1. Zhou's G (p. 8) and (7) (p. 9) against the chord display in the proof of Theorem 1.1, p. 20: integrated, it gives $G(a,b) > 0$ for $\Gamma = q$, while the four relations give $G(a,b) \le 0$.
2. Zhou's Γ''' (p. 15) and Lopatto's Lemma 2.5 (p. 6) against Lemma 2.3 and its proof, pp. 7–8.
3. Chen 2026, Prop. 4.3 (pp. 22–23; aiXiv, Prop. 4.2) against the proof of Theorem 1.1, p. 20.

F3. The Cole–Hopf slope invariant as sign inequalities for the backward drift

Summary. Lopatto's Proposition 3.6, which he calls the Cole–Hopf slope invariant, shows that every finite Cole–Hopf cascade from $\log\cosh$ satisfies $K \ge 0$ and four further inequalities, where $KB = x_r$ is the speed of the level sets of the slope $B = u_x$ [4, pp. 15–18]; Chen's paper transports them to zero temperature by smoothing $|x|$ to $\lambda^{-1}\log\cosh\lambda x$ and rescaling (proof of Prop. 3.11), and proves $u_{xxx} \le 0$ by a maximum principle (Lemma 3.4) [1], both already in its aiXiv version [2]. The preprint cites neither where it uses this content: it mentions Lopatto's invariant only in its introduction, in describing his proof, and cites Chen's Proposition 3.11 only there, for "smooth-terminal approximation" (§1.1, p. 3); its §3 cites Lopatto only for the density transform (p. 12), and Chen's Lemma 3.4 nowhere. Lemma 3.2, p. 11 states $w \le 0$, $H = r_\tau/r \le 0$ and $H_x \ge 0$ on $x > 0$: *same result*, because in Lopatto's variables (the table below; not the preprint's certificate B, K) $H = -CK$, $H_x = -C(CK)_B$ and $w = cCC_B = -2cC(c+K)B$, so $H \le 0$ is his $K \ge 0$, $w \le 0$ follows from it (it is his $C_B \le 0$), and $H_x \ge 0$ follows from his inequalities in a few lines. The approach is Lopatto's: follow the sign of K through a finite cascade started at $\log\cosh$, where $K = 0$, using that each Cole–Hopf step preserves it (by a maximum principle for $J = (BK)_B$ and the average (3.61)) and that lowering the parameter from a to b adds $a - b$ to K; the preprint's maximum principles for H and H_x in x are its own, and its proof of $w \le 0$ follows Chen's paper.

Correspondence.

	Preprint, §3	Lopatto [4]; Chen [1]
Data	$0 < c_1 \le \cdots \le c_N \le M$, terminal $M^{-1}\log(2\cosh Mx)$ (p. 8); on a step, $\gamma = c$	cascades from $(\log\cosh, 1)$ (Prop. 3.6, p. 15); $a = c$, $u = \Phi$, $B = r/c$, $C = v/c$, his time r is τ; Chen (3.47), p. 15: after $y = \lambda x$, coefficients $\le \lambda$ from $\lambda^{-1}\log\cosh\lambda x$ give such cascades
Drift	$H = r_\tau/r = w/(2r) + v$ (3.8), p. 11	$x_r = KB$ (3.44), p. 16, and $C_B = -2(a+K)B$ (3.39), p. 15, so $H = -CK$
Signs	$w \le 0$, $H \le 0$, $H_x \ge 0$ on $x > 0$ (3.8); $w \le 0$ by a maximum principle (p. 11)	$K \ge 0$ (3.41), p. 16; $C_B \le 0$ (p. 17); Chen: $u_{xxx} \le 0$ (3.20), p. 10, by the same maximum principle, (B.11), p. 32
Steps	terminal $H = (cM - M^2)\operatorname{sech}^2(Mx)$; jumps $H_{\text{new}} = H_{\text{old}} - (1-\alpha)v_{\text{old}}$, $(H_x)_{\text{new}} = (H_x)_{\text{old}} - (1-\alpha)w_{\text{old}}$, $\alpha = c_-/c_+$ (pp. 11–12)	$K = J = 0$ at the start (3.67); $K_b = K_a + a - b$ (3.65), p. 18; lowering M to c gives $K = M - c$, and (3.65) gives the jump rules

How to check.

1. Put $r = cB$, $v = cC$, $w = cCC_B$ into (3.8) and use [4, (3.38), (3.39), (3.44), pp. 15–16] to get $H = -CK$. For $H_x \geq 0$: with his $z = -\frac{1}{2}C_B$ (3.42), $\varphi := -B(CK)_B$ vanishes at $B = 0$, and $\varphi_B = BK(a + 2J - K) + zBK_B + CK_B + (3zJ - CJ_B) \geq 0$ by (3.41), (3.43).
2. Terminal H and jumps (pp. 11–12) against [4, (3.65), (3.67), p. 18], scaled by [1, (3.47), p. 15].
3. The proof of $w \leq 0$ on p. 11 against [1, Lemma 3.4, p. 10; App. B.3, (B.11), p. 32], times c.

F4. Monotonicity of the ratio of the forward score to the drift

Summary. Zhou's August 2026 paper on the Gardner transition in the Ising pure p-spin glass proves in App. E.3 that $w_0 = W_y/B$ (in the table's variable y, Zhou's x), the lower-history part of his density score divided by the slope B, is nondecreasing on $y > 0$, since $R = W_{yy}/C$ is [5, (E.37), (E.41), p. 128]; the preprint does not cite this paper, and cites Zhou only for interval support near the critical temperature [3] (§1.1, p. 3). It states the ratio sign $j = r\,\partial_x(s/r) \geq 0$, with its invariance at forward jumps, in Lemma 3.3, p. 12, as "the additional ratio sign needed by the coercive identity" (§3.3), and proves it on p. 13 from $a \geq 1/t$ and $s_{xx} \leq 0$ (Zhou's $W_{yy} \geq 0$ and $-W_{yyy} \geq 0$, (E.31)) and the backward signs of Lemma 3.2. *same result*: after the table's rescaling, $s/r = y/(\theta mB) + w_0/m$, and $B > yC$ [5, p. 130] makes the first term nondecreasing. Both need such a sign for forward-density terms of a third-derivative identity on a constant interval (for Zhou, the covariance identity (E.27), which the sign enters through (E.45); in the preprint, Lemma 3.3 → Lemma 3.4 → Lemma 4.1 and Proposition 4.2 → Proposition 5.1 → Theorem 1.1), and both use Zhou's insight of measuring the score against the drift: a jump adds a multiple of the drift to the score, so the ratio's monotonicity survives jumps, and on the Gaussian first step it is concavity of the drift. Between jumps Zhou propagates $R_y \geq 0$ and averages, while the preprint propagates $\rho = s_x/s - v/r \geq 0$, each by a maximum principle; Zhou's p. 3 says his appendix proofs were drafted by language models, pending his revision.

Correspondence.

	Preprint [10]	Zhou [5, App. E, pp. 124–130]
Data	$0 < c_1 \leq \cdots \leq c_N \leq M$; terminal datum $M^{-1}\log(2\cosh Mx)$; $X_0 = 0$ (§3, pp. 8–9)	masses $m = c/M \in (0,1]$, atoms $d = \delta/M$; $W = 0$ before the first atom (E.28); upper cone (E.52) from $\log\cosh y$ (E.46), $J = 1 - m \geq 0$ (E.50), also for $m = 1$
Dictionary	$r = c\Phi_x$, $v = r_x$; $f = e^{c\Phi}g$, with g a heat solution between jumps and $g_+ = g_- e^{-\delta\Phi}$ at a jump $\delta = c_+ - c_-$ (3.2), p. 9; $s = -\partial_x \log g$, $a = s_x$	$\theta = M^2 t$, $y = Mx$ (Zhou's t, x); $U = M\Phi - \log 2$ solves (E.1); $B = U_y$, $C = U_{yy}$ (E.2), not the B of Section 2; $g \propto pe^{-mU} \propto e^{-y^2/(2\theta)-W}$ (E.4)–(E.5), p the density of MX_t; $r = MmB$, $v = M^2 mC$, $s = x/t + MW_y$, $a = 1/t + M^2 W_{yy}$
Ratio sign	$j = a - (s/r)v = r\,\partial_x(s/r) \geq 0$ (3.11), p. 12; first step: $s = x/t$ and $r \geq xv$; later steps: maximum principle for $\rho = s_x/s - v/r$ (3.16)	$s/r = y/(\theta mB) + w_0/m$, so $j = M^2[(B - yC)/(\theta B) + C(R - w_0)]$ on $y > 0$, with $B > yC$ (p. 130) and $R \geq w_0$ (E.41), from $R_y \geq 0$ (E.37), which uses (E.52); first step: $W = 0$
Jumps	$j_+ = j_-$ by cancellation (p. 13), with $s_+ = s_- + \delta\Phi_x$ (3.15), p. 12	the atom $W \mapsto W + dU$ (E.28) gives $R \mapsto R + d$ and $w_0 \mapsto w_0 + d$; B, C continuous

How to check.

1. Check the dictionary (§3, pp. 8–9; Zhou's (E.1)–(E.5), (E.28) [5, pp. 124–127]), derive j from (3.11), (E.32) and (E.40), and read off its signs from (E.41) and $B > yC$ (p. 130).
2. Compare the first step and jump on p. 13 with (E.28) and the line after (E.36) [5, pp. 127–128].

F5. Excluding a terminal plateau by a square-root boundary layer

Summary. H.-B. Chen's paper [1, Prop. 4.5, pp. 23–25] proves that the zero-temperature minimizer has no constant stretch ending at 1, first in its aiXiv predecessor of July 16, 2026 [2, Prop. 4.4, pp. 32–33], which it credits [1, p. 1; App. C, p. 38]; the preprint cites Chen's paper only for other results and the aiXiv version nowhere. The preprint proves the same statement as Lemma 6.3, pp. 19–20, which settles the terminal and empty-support cases of the proof of Theorem 1.1, p. 20, and gives the proof as its own (p. 19; §1.1, p. 3), crediting only the conclusion to Auffinger, Chen and Zeng [6, Remark 7, p. 15], who assert it and indicate a different, perturbative route without details. Both proofs run a boundary layer at $t = 1$: the Cole–Hopf formula makes $u(1-\epsilon, \sqrt{\epsilon}\, y)$ a sign average under the tilt $e^{c\sqrt{\epsilon}|y+Z|}$, harmless at this scale, so $1 - u^2$ is of order one on a window of width $\sqrt{\epsilon}$ where X_t has density of order one; then $1 - q(t)$ is at least of order $\sqrt{1-t}$, and $G < 0$ near 1 contradicts $G \geq 0$. The case $c = 0$ goes back to Chen, Handschy and Lerman [8, proof of Thm. 6, pp. 14–16], for $\gamma \equiv 0$, and to El Alaoui, Montanari and Sellke [7]; the extension to a positive plateau, which behaves like $c = 0$ in this layer, first appears in the aiXiv version of Chen's paper. *same result*: as $0 \in S$ (Chen's Prop. 3.2), S is closed and the minimizer unique [8, Thm. 4, p. 8], every terminal plateau lies in one starting at $\max S \in S$, so Chen's Proposition 4.5 is Lemma 6.3; the preprint replaces Chen's limits (4.30) and (4.33), and the asymptotic (4.38), by fixed-window lower bounds adapted from the El Alaoui–Montanari–Sellke argument, which it calls related (p. 19).

Correspondence. Here $\epsilon = 1 - t$, $h(t, y) = \mathbb{E}e^{c|y+\sqrt{1-t}\,Z|}$, p_v is the $N(0, v)$ density, ρ_s the law of X_s ($s \in (a, 1)$ fixed); Chen's u is the preprint's Φ, and Chen's ρ_t the density of X_t.

	Preprint, Lemma 6.3	Chen [1]
Statement	for a minimizer, no $a < 1$ and finite $c \geq 0$ with $\gamma = c$ on $(a, 1)$ (p. 19)	for the minimizer $\gamma_\star$, no $a < 1$ with $(a, 1) \cap S = \varnothing$ and $a \in S$, Prop. 4.5, p. 23; $0 \in S$, Prop. 3.2, p. 8
Gradient at scale $\sqrt{\epsilon}$	for $\lvert y\rvert \leq \sqrt{\epsilon}$, $u(t, y) = \mathbb{E}[\operatorname{sgn}(y + \sqrt{\epsilon}Z)e^{c\lvert y+\sqrt{\epsilon}Z\rvert}]/\mathbb{E}e^{c\lvert y+\sqrt{\epsilon}Z\rvert}$ (p. 20)	$\mathsf{B}_\varepsilon(y) := u_x(1-\varepsilon, \sqrt{\varepsilon}\, y) = \mathbb{E}[\operatorname{sign}(y + Z)e^{m\sqrt{\varepsilon}\lvert y+Z\rvert}]/\mathbb{E}e^{m\sqrt{\varepsilon}\lvert y+Z\rvert}$, (4.28)–(4.29), p. 23
Law of X_t on the plateau	$k_{s,t}(x, y) = p_{t-s}(y - x)h(t, y)/h(s, x)$ (6.6), p. 19; its mixture against ρ_s is bounded below on $\lvert y\rvert \leq 1$ (6.7), p. 20	$Q = \rho_t e^{-mu}$ solves the heat equation, (3.21)–(3.22), p. 10; $\rho_{1-\varepsilon}(\sqrt{\varepsilon}\, y) \to \rho_1(0) > 0$, (4.33), p. 24
Layer and contradiction	$1 - q(t) \geq C\sqrt{1-t}$ (6.5), p. 19; $G < 0$ near 1, against Lemma 6.2 (p. 20)	$1 - \Gamma(t) \sim (4/\sqrt{\pi})\, \rho_1(0)\sqrt{1-t}$, (4.38), and $G < 0$ near 1, against (3.8), p. 25

How to check.

1. The display for $u(t, y)$ (p. 20) after $y \mapsto \sqrt{\epsilon}\, y$ against [1, (4.28)–(4.29), p. 23] with $m = c$; (6.6) against [1, (3.21)–(3.22), p. 10] with $e^{mu} = h$.

2. The end of the proof of Lemma 6.3 (p. 20) against [1, (4.35)–(4.38) and Step 4, pp. 24–25], and the whole proof against [2, (4.19)–(4.25), pp. 32–33]; (6.7) and the bound $1 - u^2 \geq \kappa$ against [7, Lemma 6.12, p. 29, and the end of the proof of Thm. 5, p. 31], where $c = 0$.

F6. The origin lies in the support: no initial interval of zero mass

Summary. Chen, Handschy and Lerman (CHL) proved in 2016 that at zero external field the smallest point of the support of the zero-temperature Parisi measure is 0 [8, §3.2, p. 18], for the mixed even p-spin models, SK with $\xi(s) = s^2/2$ among them. The preprint cites their paper only for other results, such as uniqueness and the variational conditions (§1.1, p. 3; pp. 5, 18, 20), but not for this result, which it proves again without a citation as the initial-component case of the proof of Theorem 1.1, p. 20. The argument it uses, and the insight behind it, are Auffinger and Chen's [9, proof of Thm 1(i), p. 15] for the positive-temperature analogue $0 \in \operatorname{supp} \mu_P$: on an initial interval of zero mass the slope of the self-consistency function increases strictly, so the function, which vanishes at 0 by symmetry, cannot climb to the diagonal at a positive first support point b, where the contact condition bounds its slope by 1; the preprint cites that paper only for its moment method (pp. 3, 20). *same result*: CHL's statement is the conclusion of the step, for the same functional and class, and their overlap map ψ_1 is q on $[0, q_h]$ (below), so they reach the same two-fixed-point contradiction, which they close with W.-K. Chen's fixed-point uniqueness theorem [11, Thm 5] rather than with a slope bound on the gap. H.-B. Chen's 2026 paper [1, Prop. 3.2, pp. 8–9], like its aiXiv version [2, p. 17], states $0 \in S$ and takes it from CHL, and El Alaoui, Montanari and Sellke [7, proof of Thm 5, pp. 30–31] run the same contradiction over a larger, non-monotone class; the preprint cites H.-B. Chen's paper (not its aiXiv version) and that proof only for other content (the proof for its terminal segment, p. 19).

Correspondence. Here $b = \min S$ if $0 \notin S$, and q_h is the smallest support point of CHL's minimizer γ_h. At $h = 0$ their overlap map is $\psi_1(s) = \mathbb{E}\, \partial_x \Phi_{\gamma_h}(q_h, \chi_1)\, \partial_x \Phi_{\gamma_h}(q_h, \chi_2)$, $\mathbb{E}\chi_i^2 = \xi'(q_h)$, $\mathbb{E}\chi_1\chi_2 = \xi'(s)$. As $u(t, B_t) = \mathbb{E}[u(b, B_b) \mid B_t]$ on $[0, b]$, $q(s) = \mathbb{E}[u(b, \chi_1)u(b, \chi_2)]$ with $\operatorname{Var} \chi_i = b$, $\operatorname{Cov}(\chi_1, \chi_2) = s$, which is $\psi_1(s)$ when $\xi'(s) = s$ and $q_h = b$.

	Preprint	CHL [8]	Auffinger–Chen [9]
Statement	$0 \in S$, any minimizer (p. 20)	$q_h = 0$ at $h = 0$ (p. 18; also p. 4)	$0 \in \operatorname{supp} \mu_P$, positive temperature (Thm 1(i), p. 4)
Gap ends	$q(0) = 0$ (Lemma 2.2, p. 7); $q(b) = b$, $q'(b) \leq 1$ by (6.4), p. 19	$\psi_1(0) = 0$ (Lemma 4, p. 11); $\psi_1(q_h) = q_h$, $\xi''(q_h)\mathbb{E}(\partial_{xx}\Phi_{\gamma_h})^2 \leq 1$ (34)–(35), p. 18	$\Gamma = \Gamma_{\mu_P}$ of (18), p. 8: $\Gamma(0) = 0$; $\Gamma(q_m) = q_m$, $\Gamma'(q_m) \leq 1$ (Thm 5, p. 9; p. 15)
Closing step	$(q')' = \mathbb{E}\Phi_{xxx}^2 > 0$ on $(0, b)$ (Lemma 2.3, p. 8), so $q' < q'(b) \leq 1$ and $q(b) < b$	two fixed points 0, q_h of ψ_1, against W.-K. Chen's Thm 5	$\Gamma'' = \gamma_{1,\mu_P} > 0$ on $(0, q_m)$ (Prop. 3, pp. 8–9): one fixed point on $[0, q_m]$

How to check.

1. The initial-component case (p. 20) and the $c = 0$ clause of Lemma 2.3 (proof on p. 8) against [9, proof of Thm 1(i), p. 15] and [9, Prop. 3, (20)–(22), pp. 8–9] with $\mu([0, u]) = 0$, $\xi''' = 0$.

2. The same step against [8, §3.2, p. 18]: (34)–(35) are $q(b) = b$, $q'(b) \leq 1$ of (6.4); and [8, (5)–(6), Thm 4, p. 8] at $h = 0$ against (1.2)–(1.3).
3. [1, end of the proof of Prop. 3.2, p. 9] and [2, p. 17]: $0 \in S$, taken from CHL.

F7. Full support verifies the no-overlap-gap assumption of the optimization algorithm

Summary. Chen's paper [1, §1, p. 4] already observes that its zero-temperature full-support theorem verifies, for SK, the no-overlap-gap assumption of El Alaoui, Montanari and Sellke, under which their message-passing algorithm comes within any $\epsilon > 0$ of the optimum. The preprint cites Chen's paper only for the full-support theorem and proof steps: its Corollary 7.3, p. 25 and the announcements of it on pp. 2, 3 and 25 name or cite only El Alaoui, Montanari and Sellke, for the assumption and the corollary. The shared approach is one deduction: full support makes the minimizer strictly increasing on $[0, 1)$, which is exactly their Assumption 2 for SK, so their Corollary 2.2 applies; they posed the assumption for this use and expected it to hold for SK [7, p. 8], and the observation that the zero-temperature full-support theorem settles it for SK is already in Chen's paper (the theorem itself is finding F1). The relationship is *same result*, since both papers draw the same conclusion by this deduction; to apply Corollary 2.2, the preprint adds an independent diagonal that puts H_n in its Gaussian input format, carrying over the guarantee and the cost, and takes the operation count $C(\epsilon)n^2$ from Theorem 3 of El Alaoui, Montanari and Sellke. Corollary 7.3 feeds no other result of the preprint, hence *peripheral*.

Correspondence.

	Preprint	Chen [1]
Strict increase	$\operatorname{supp} \mu_\gamma = [0, 1)$, so $\gamma(b) > \gamma(a)$ for $0 \leq a < b < 1$ (Theorem 1.1, p. 2)	$\operatorname{supp}_{[0,1)} \nu_\star = [0, 1)$ (Theorem 1.3, p. 3), so $\gamma_\star(t) - \gamma_\star(s) = \nu_\star((s, t]) > 0$ for $0 \leq s < t < 1$ (p. 4)
Assumption	Assumption 2 of El Alaoui et al. for $\xi(t) = t^2/2$ (proof of Corollary 7.3)	Assumption 2 of El Alaoui et al. for SK (p. 4)
Conclusion	their Corollary 2.2: $\Pr\{H_n(\sigma^{\mathrm{alg}}) \geq \max_\sigma H_n(\sigma) - \epsilon n\} \to 1$ (Corollary 7.3)	their Corollary 2.2: energy within any fixed $\varepsilon > 0$ of the limiting optimum (p. 4)

How to check.

1. The proof of Corollary 7.3, p. 25 against [1, p. 4], the paragraph beginning "Full support is also relevant to optimization algorithms," and both against [7, Assumption 2, Corollary 2.2 and (2.8), p. 8].
2. The paragraphs on pp. 2–3 and §7.1, pp. 25–26: Chen's paper is cited for Theorem 1.3 and for proof steps, but not for the algorithmic application.

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